

YAROL'IN, Russian Federation

WMO: 242190

Lat: 67.133N

Lon: 108.533E

Elev: 236

StdP: 98.52

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-54.2	-52.2	-55.5	0.0	-53.8	-53.5	0.0	-52.2	9.3	-13.4	7.9	-14.2	0.2	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	27.9	17.9	25.5	16.8	23.1	15.3	18.7	26.0	17.3	24.1	16.0	21.9	2.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	11.8	21.5	14.5	10.6	19.9	13.2	9.7	18.5	54.4	26.1	49.6	24.0	45.6	22.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.2	7.0	6.1	DB	-54.9	31.1	2.1	2.5	-56.4	32.9	-57.6	34.4	-58.7	35.8	-60.2	37.6	(4)
(5)			WB	-54.9	20.6	2.1	2.4	-56.4	22.3	-57.6	23.7	-58.7	25.0	-60.2	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.6	-38.0	-31.8	-19.9	-10.1	0.8	12.7	14.5	11.4	3.6	-8.9	-28.2	-34.2
	DBStd	20.15	10.42	10.42	9.53	7.18	5.79	5.68	4.14	3.91	4.70	7.73	9.02	10.44
	HDD10.0	7851	1487	1171	926	602	288	36	8	30	199	587	1146	1371
	HDD18.3	10578	1746	1404	1184	852	543	184	132	218	442	846	1396	1629
	CDD10.0	346	0	0	0	0	3	116	148	72	6	0	0	0
	CDD18.3	31	0	0	0	0	0	15	14	2	0	0	0	0
	CDH23.3	478	0	0	0	0	0	228	202	44	4	0	0	0
	CDH26.7	115	0	0	0	0	0	67	42	6	0	0	0	0
Wind	WSAvg	2.2	1.5	1.6	2.3	2.9	2.6	2.6	2.1	2.2	2.4	2.3	1.7	1.7
Precipitation	PrecAvg	388	14	11	15	22	34	55	53	63	41	38	24	18
	PrecMax	534	25	25	26	42	59	89	85	125	73	54	41	45
	PrecMin	284	4	3	5	8	15	14	14	22	9	18	11	6
	PrecStd	51	4	5	6	7	12	17	18	23	16	11	7	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.2	-6.3	4.2	7.6	19.7	31.2	29.8	27.5	23.5	7.2	-5.0	-5.2
		MCWB	-8.0	-7.5	1.5	2.8	11.5	19.0	18.3	17.4	14.9	4.6	-5.8	-6.5
	2%	DB	-11.6	-9.7	1.3	4.9	16.0	28.6	27.8	24.1	16.5	4.7	-8.8	-9.6
		MCWB	-12.3	-10.6	-1.0	1.4	9.9	17.6	18.6	15.7	10.6	2.6	-9.7	-10.6
	5%	DB	-16.0	-11.7	-1.6	3.2	12.6	26.1	25.5	21.5	13.2	2.7	-11.8	-12.6
		MCWB	-16.4	-12.6	-3.3	0.7	7.3	16.1	17.3	14.5	8.8	1.1	-12.3	-13.3
	10%	DB	-20.4	-15.5	-4.6	1.1	9.7	23.0	23.3	18.8	10.7	0.7	-14.7	-17.1
		MCWB	-20.8	-16.0	-6.1	-1.1	5.6	15.1	15.5	13.3	7.1	-0.7	-15.2	-17.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.1	-7.3	1.6	3.6	11.8	20.2	21.4	18.6	14.9	4.6	-6.0	-6.6
		MCDB	-7.0	-6.3	3.5	6.5	18.8	28.8	26.8	25.2	23.2	6.6	-5.4	-5.2
	2%	WB	-12.3	-10.5	-1.0	1.9	9.9	18.4	19.5	16.9	11.1	2.8	-9.7	-10.5
		MCDB	-11.6	-9.7	1.1	4.9	15.4	26.5	25.9	22.7	15.4	4.5	-8.9	-9.7
	5%	WB	-16.6	-12.5	-3.4	0.7	7.8	17.0	17.7	15.4	9.2	1.2	-12.3	-13.4
		MCDB	-16.1	-11.6	-1.7	3.1	11.7	24.3	24.3	20.0	12.5	2.7	-11.8	-12.6
	10%	WB	-20.7	-16.0	-6.1	-1.0	5.7	15.5	16.3	14.0	7.5	-0.8	-15.1	-17.6
		MCDB	-20.3	-15.6	-4.8	1.0	9.6	22.4	22.1	17.7	10.5	0.6	-14.8	-17.2
Mean Daily Temperature Range		MDBR	8.4	11.1	14.7	13.4	9.1	10.7	11.3	10.2	8.1	7.8	9.0	9.0
	5% DB	MCDBR	11.6	11.0	12.1	13.2	12.8	15.6	15.7	16.3	13.4	6.4	11.7	13.2
		MCWBR	11.0	10.7	10.6	10.8	7.8	6.9	7.9	8.9	8.3	4.8	11.0	12.6
	5% WB	MCDBR	11.4	11.2	12.2	12.4	12.3	14.2	13.8	13.2	12.1	6.4	11.8	13.3
		MCWBR	11.0	10.8	10.8	10.3	7.9	7.0	7.5	7.7	8.1	5.1	11.3	12.6
Clear-Sky Solar Irradiance	taub	0.193	0.240	0.291	0.320	0.348	0.364	0.430	0.413	0.313	0.249	0.182	N/A	
	taud	1.860	2.024	2.047	2.073	2.195	2.407	2.195	2.194	2.455	2.164	1.886	N/A	
	Ebn at Noon	147	578	738	822	841	841	755	720	746	568	146	0	
	Edh at Noon	13	54	92	119	118	100	120	108	64	45	11	0	
All-Sky Solar Radiation	RadAvg	0.06	0.71	2.38	4.53	5.69	5.49	4.93	3.30	1.85	0.78	0.15	0.00	
	RadStd	0.01	0.04	0.14	0.26	0.36	0.51	0.43	0.43	0.20	0.09	0.01	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page